POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6034B | Semester VI | Theory | 4 Credits

Energy Conservation and Management

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6034B as Energy Conservation and Management in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Sustainable Energy Technology course and polytechnic energy management curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Energy policies, DSM programs, building designs and sustainability plans must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support energy management principles, DSM, green buildings, and global sustainability goals. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Energy Conservation and Management studies the strategic and organizational approach to achieving energy efficiency and sustainability. It develops the ability to formulate energy policies, implement demand-side management (DSM) techniques, evaluate green building designs, and understand global energy governance and sustainability goals while respecting the technical and safety boundaries of electrical and environmental engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Basic electrical engineering, energy conservation and audit, and industrial management.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the principles of energy management, energy policy and the role of energy managers.
2. Apply demand-side management (DSM) techniques to optimize utility load profiles.
3. Analyze sustainable energy technologies and green building concepts including life cycle assessment.
4. Explain global energy trends, carbon trading and the UN Sustainable Development Goals (SDGs).

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain energy management principles and formulate organizational energy policies.	Understand
CO2	Implement demand-side management (DSM) strategies for load optimization.	Apply
CO3	Evaluate green building designs and perform basic life cycle assessments.	Apply
CO4	Analyze global energy governance, carbon trading and sustainability goals.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Energy Management Principles and Policy	Definition and objectives of energy management; Energy policy: National and organizational; Energy manager: roles and responsibilities; Energy security and its importance; Monitoring and Targeting (M&T) systems.	Approx. 14 hours	CO1	Module 1
2	Demand Side Management (DSM)	Concept and benefits of DSM; DSM techniques: Peak clipping, Valley filling, Load shifting, Strategic conservation; Role of utilities in DSM; Time-of-Use (ToU) pricing; Load management systems.	Approx. 16 hours	CO2	Module 2
3	Sustainable Energy and Green Buildings	Introduction to sustainability; Life Cycle Assessment (LCA) basics; Green building concepts: LEED, GRIHA ratings; Passive solar design; Energy efficient building materials; Net Zero Energy Buildings (NZEB).	Approx. 16 hours	CO3	Module 3
4	Global Trends and Energy Governance	Global energy trends; Carbon footprint and Carbon trading; Renewable Energy Certificates (RECs); International energy treaties; UN Sustainable Development Goals (SDGs) related to energy.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Energy Management Principles and Policy

Definition and objectives of energy management; Energy policy: National and organizational; Energy manager: roles and responsibilities; Energy security and its importance; Monitoring and Targeting (M&T) systems.

CO1 · Approx. 14 hours

Demand Side Management (DSM)

Concept and benefits of DSM; DSM techniques: Peak clipping, Valley filling, Load shifting, Strategic conservation; Role of utilities in DSM; Time-of-Use (ToU) pricing; Load management systems.

CO2 · Approx. 16 hours

Sustainable Energy and Green Buildings

Introduction to sustainability; Life Cycle Assessment (LCA) basics; Green building concepts: LEED, GRIHA ratings; Passive solar design; Energy efficient building materials; Net Zero Energy Buildings (NZEB).

CO3 · Approx. 16 hours

Global Trends and Energy Governance

Global energy trends; Carbon footprint and Carbon trading; Renewable Energy Certificates (RECs); International energy treaties; UN Sustainable Development Goals (SDGs) related to energy.

CO4 · Approx. 14 hours

MODULE 1

Energy Management Principles and Policy

Definition and objectives of energy management; Energy policy: National and organizational; Energy manager: roles and responsibilities; Energy security and its importance; Monitoring and Targeting (M&T) systems.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

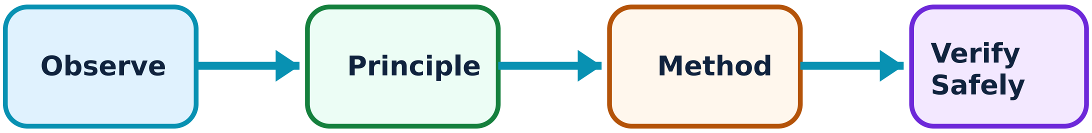
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Management Principles and Policy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Management and Policy−

Energy management is the strategy of adjusting and optimizing energy use to reduce requirements per unit of output. An organizational energy policy provides the commitment and framework for energy saving. Energy security ensures the continuous availability of energy at an affordable price.

Study and application method

List the key elements of an 'Organizational Energy Policy'; explain the 'Plan-Do-Check-Act' (PDCA) cycle in energy management.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the key elements of an 'Organizational Energy Policy'; explain the 'Plan-Do-Check-Act' (PDCA) cycle in energy management.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Energy policies must align with local laws. Do not implement policy changes without authorised legal review and stakeholder consultation.

2. Monitoring and Targeting (M&T)−

M&T is a management technique that uses energy information as a basis to eliminate waste, reduce current levels of energy use, and improve existing operating procedures. It involves establishing energy baselines and setting realistic saving targets.

Study and application method

Describe the four steps of the 'M&T' loop; explain the significance of an 'Energy Baseline' in monitoring performance.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the four steps of the 'M&T' loop; explain the significance of an 'Energy Baseline' in monitoring performance.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Target setting requires accurate historical data. Do not set energy targets without authorised verification of meter data and production variables.

Energy-use calculator

Appliance power (W)

100

Hours used

5

Tariff per unit

6

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Demand Side Management (DSM)

Concept and benefits of DSM; DSM techniques: Peak clipping, Valley filling, Load shifting, Strategic conservation; Role of utilities in DSM; Time-of-Use (ToU) pricing; Load management systems.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

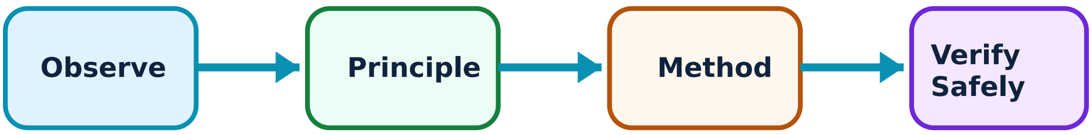
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Demand Side Management (DSM): identify the system, state its principle, follow the correct method and verify the result safely.

1. DSM Concepts and Techniques−

Demand Side Management involves actions taken on the customer's side of the meter to change the amount or timing of energy consumption. Techniques like 'Load Shifting' move consumption from peak to off-peak hours, while 'Peak Clipping' reduces demand during peak periods.

Study and application method

Compare 'Peak Clipping' and 'Load Shifting' with conceptual load curves; explain the benefits of DSM for both the utility and the consumer.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Peak Clipping' and 'Load Shifting' with conceptual load curves; explain the benefits of DSM for both the utility and the consumer.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Load management can affect industrial processes. Do not implement load shifting without authorised analysis of the impact on production quality and safety.

2. Utility DSM and Pricing−

Utilities use DSM to avoid building new power plants and to improve grid stability. Time-of-Use (ToU) pricing encourages consumers to use electricity during off-peak hours by offering lower rates. Load management systems use smart meters and controllers to automate DSM.

Study and application method

Explain the concept of 'Time-of-Use' (ToU) pricing; describe the role of 'Smart Meters' in implementing DSM.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the concept of 'Time-of-Use' (ToU) pricing; describe the role of 'Smart Meters' in implementing DSM.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: DSM programs involve financial incentives and contracts. All utility-side DSM measures must follow authorised regulatory frameworks and consumer protection codes.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Sustainable Energy and Green Buildings

Introduction to sustainability; Life Cycle Assessment (LCA) basics; Green building concepts: LEED, GRIHA ratings; Passive solar design; Energy efficient building materials; Net Zero Energy Buildings (NZEB).

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

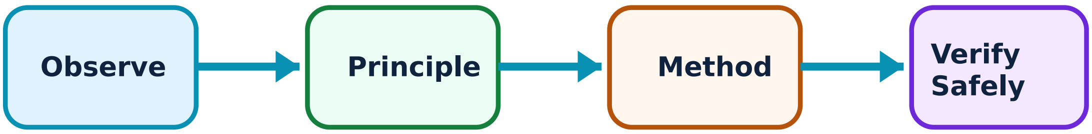
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sustainable Energy and Green Buildings: identify the system, state its principle, follow the correct method and verify the result safely.

1. Sustainability and LCA−

Sustainability meets the needs of the present without compromising future generations. Life Cycle Assessment (LCA) evaluates the environmental impact of a product or building from 'cradle to grave', including material extraction, use, and disposal.

Study and application method

Define 'Sustainability' in the context of energy; explain the stages of a 'Life Cycle Assessment' (LCA).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Sustainability' in the context of energy; explain the stages of a 'Life Cycle Assessment' (LCA).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: LCA results depend on the boundary conditions and data quality. Do not use LCA for commercial claims without authorised peer review and adherence to ISO standards.

2. Green Building Technologies−

Green buildings minimize environmental impact through efficient use of energy, water, and materials. Rating systems like LEED and GRIHA provide certification. Passive solar design uses building orientation and windows to manage heat and light naturally.

Study and application method

List five features of a 'Green Building'; explain the concept of a 'Net Zero Energy Building' (NZEB).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List five features of a 'Green Building'; explain the concept of a 'Net Zero Energy Building' (NZEB).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Green building certifications require rigorous documentation. Do not claim certification levels without authorised audit by the respective rating agency.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Global Trends and Energy Governance

Global energy trends; Carbon footprint and Carbon trading; Renewable Energy Certificates (RECs); International energy treaties; UN Sustainable Development Goals (SDGs) related to energy.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

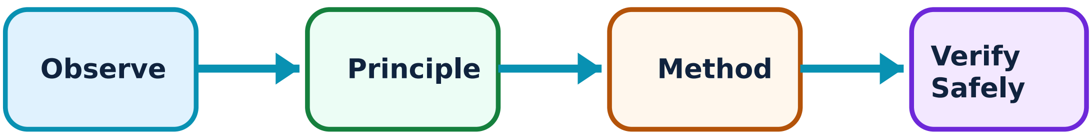
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Global Trends and Energy Governance: identify the system, state its principle, follow the correct method and verify the result safely.

1. Carbon Trading and RECs−

Carbon trading allows organizations to buy and sell 'Carbon Credits' to meet emission targets. Renewable Energy Certificates (RECs) represent the environmental attributes of 1 MWh of renewable energy, allowing utilities to meet renewable purchase obligations (RPO).

Study and application method

Explain the principle of 'Cap and Trade' in carbon markets; describe the purpose of 'Renewable Energy Certificates' (RECs).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the principle of 'Cap and Trade' in carbon markets; describe the purpose of 'Renewable Energy Certificates' (RECs).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ▯▯▯▯▯ concept ▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯ diagram / circuit / formula / procedure ▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯▯▯▯▯▯ result ▯▯▯▯▯▯▯▯▯▯▯ application ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Common mistake / precaution: Carbon markets are complex and volatile. Do not provide investment advice on carbon credits without authorised financial and environmental risk analysis.

2. International Governance and SDGs−

International treaties like the Paris Agreement set global emission targets. The UN SDG 7 aims to 'ensure access to affordable, reliable, sustainable and modern energy for all'. Global energy governance involves cooperation between nations to ensure energy security and sustainability.

Study and application method

List the targets of 'UN SDG 7'; explain the significance of the 'Paris Agreement' in global energy policy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the targets of 'UN SDG 7'; explain the significance of the 'Paris Agreement' in global energy policy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: International energy agreements are subject to geopolitical changes. All global trend analyses must follow authorised diplomatic and environmental reporting standards.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Energy Management Principles and Policy

Use the concept flow to revise observation, principle, method and verification.

Module 2: Demand Side Management (DSM)

Use the concept flow to revise observation, principle, method and verification.

Module 3: Sustainable Energy and Green Buildings

Use the concept flow to revise observation, principle, method and verification.

Module 4: Global Trends and Energy Governance

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Draft a simple 'Energy Policy Statement' for your college or a small business.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare ToU pricing and Flat-rate pricing in a conceptual table showing consumer impacts.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the 'GRIHA' or 'LEED' rating of a prominent building in your region.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate your personal 'Carbon Footprint' conceptually using an online calculator or simple data.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Research and list the 'Renewable Purchase Obligations' (RPO) for your state utility.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Energy Management Principles and Policy

Explain Management and Policy and state one correct method, application or precaution.−

Energy management is the strategy of adjusting and optimizing energy use to reduce requirements per unit of output. An organizational energy policy provides the commitment and framework for energy saving. Energy security ensures the continuous availability of energy at an affordable price.

Answer framework: List the key elements of an 'Organizational Energy Policy'; explain the 'Plan-Do-Check-Act' (PDCA) cycle in energy management.

Important point: Energy policies must align with local laws. Do not implement policy changes without authorised legal review and stakeholder consultation.

Explain Monitoring and Targeting (M&T) and state one correct method, application or precaution.−

M&T is a management technique that uses energy information as a basis to eliminate waste, reduce current levels of energy use, and improve existing operating procedures. It involves establishing energy baselines and setting realistic saving targets.

Answer framework: Describe the four steps of the 'M&T' loop; explain the significance of an 'Energy Baseline' in monitoring performance.

Important point: Target setting requires accurate historical data. Do not set energy targets without authorised verification of meter data and production variables.

Module 2 — Demand Side Management (DSM)

Explain DSM Concepts and Techniques and state one correct method, application or precaution.−

Demand Side Management involves actions taken on the customer's side of the meter to change the amount or timing of energy consumption. Techniques like 'Load Shifting' move consumption from peak to off-peak hours, while 'Peak Clipping' reduces demand during peak periods.

Answer framework: Compare 'Peak Clipping' and 'Load Shifting' with conceptual load curves; explain the benefits of DSM for both the utility and the consumer.

Important point: Load management can affect industrial processes. Do not implement load shifting without authorised analysis of the impact on production quality and safety.

Explain Utility DSM and Pricing and state one correct method, application or precaution.−

Utilities use DSM to avoid building new power plants and to improve grid stability. Time-of-Use (ToU) pricing encourages consumers to use electricity during off-peak hours by offering lower rates. Load management systems use smart meters and controllers to automate DSM.

Answer framework: Explain the concept of 'Time-of-Use' (ToU) pricing; describe the role of 'Smart Meters' in implementing DSM.

Important point: DSM programs involve financial incentives and contracts. All utility-side DSM measures must follow authorised regulatory frameworks and consumer protection codes.

Module 3 — Sustainable Energy and Green Buildings

Explain Sustainability and LCA and state one correct method, application or precaution.−

Sustainability meets the needs of the present without compromising future generations. Life Cycle Assessment (LCA) evaluates the environmental impact of a product or building from 'cradle to grave', including material extraction, use, and disposal.

Answer framework: Define 'Sustainability' in the context of energy; explain the stages of a 'Life Cycle Assessment' (LCA).

Important point: LCA results depend on the boundary conditions and data quality. Do not use LCA for commercial claims without authorised peer review and adherence to ISO standards.

Explain Green Building Technologies and state one correct method, application or precaution.−

Green buildings minimize environmental impact through efficient use of energy, water, and materials. Rating systems like LEED and GRIHA provide certification. Passive solar design uses building orientation and windows to manage heat and light naturally.

Answer framework: List five features of a 'Green Building'; explain the concept of a 'Net Zero Energy Building' (NZEB).

Important point: Green building certifications require rigorous documentation. Do not claim certification levels without authorised audit by the respective rating agency.

Module 4 — Global Trends and Energy Governance

Explain Carbon Trading and RECs and state one correct method, application or precaution.−

Carbon trading allows organizations to buy and sell 'Carbon Credits' to meet emission targets. Renewable Energy Certificates (RECs) represent the environmental attributes of 1 MWh of renewable energy, allowing utilities to meet renewable purchase obligations (RPO).

Answer framework: Explain the principle of 'Cap and Trade' in carbon markets; describe the purpose of 'Renewable Energy Certificates' (RECs).

Important point: Carbon markets are complex and volatile. Do not provide investment advice on carbon credits without authorised financial and environmental risk analysis.

Explain International Governance and SDGs and state one correct method, application or precaution.−

International treaties like the Paris Agreement set global emission targets. The UN SDG 7 aims to 'ensure access to affordable, reliable, sustainable and modern energy for all'. Global energy governance involves cooperation between nations to ensure energy security and sustainability.

Answer framework: List the targets of 'UN SDG 7'; explain the significance of the 'Paris Agreement' in global energy policy.

Important point: International energy agreements are subject to geopolitical changes. All global trend analyses must follow authorised diplomatic and environmental reporting standards.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Energy Management Principles and Policy.

2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Demand Side Management (DSM).
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Sustainable Energy and Green Buildings.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Global Trends and Energy Governance.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and objectives of energy management; Energy policy: National and organizational; Energy manager: roles and responsibilities; Energy security and its importance; Monitoring and Targeting (M&T) systems..
2. Develop a complete answer or practical plan for: Concept and benefits of DSM; DSM techniques: Peak clipping, Valley filling, Load shifting, Strategic conservation; Role of utilities in DSM; Time-of-Use (ToU) pricing; Load management systems..
3. Develop a complete answer or practical plan for: Introduction to sustainability; Life Cycle Assessment (LCA) basics; Green building concepts: LEED, GRIHA ratings; Passive solar design; Energy efficient building materials; Net Zero Energy Buildings (NZEB)..
4. Develop a complete answer or practical plan for: Global energy trends; Carbon footprint and Carbon trading; Renewable Energy Certificates (RECs); International energy treaties; UN Sustainable Development Goals (SDGs) related to energy..

LAST-MILE REVISION

Quick Revision

Module 1: Energy Management Principles and Policy

Definition and objectives of energy management; Energy policy: National and organizational; Energy manager: roles and responsibilities; Energy security and its importance; Monitoring and Targeting (M&T) systems.

- Match each topic to CO1.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 2: Demand Side Management (DSM)

Concept and benefits of DSM; DSM techniques: Peak clipping, Valley filling, Load shifting, Strategic conservation; Role of utilities in DSM; Time-of-Use (ToU) pricing; Load management systems.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Sustainable Energy and Green Buildings

Introduction to sustainability; Life Cycle Assessment (LCA) basics; Green building concepts: LEED, GRIHA ratings; Passive solar design; Energy efficient building materials; Net Zero Energy Buildings (NZEB).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Global Trends and Energy Governance

Global energy trends; Carbon footprint and Carbon trading; Renewable Energy Certificates (RECs); International energy treaties; UN Sustainable Development Goals (SDGs) related to energy.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Management and Policy	Energy management is the strategy of adjusting and optimizing energy use to reduce requirements per unit of output.
Monitoring and Targeting (M&T)	M&T is a management technique that uses energy information as a basis to eliminate waste, reduce current levels of energy use, and improve existing operating procedures.
DSM Concepts and Techniques	Demand Side Management involves actions taken on the customer's side of the meter to change the amount or timing of energy consumption.
Utility DSM and Pricing	Utilities use DSM to avoid building new power plants and to improve grid stability.
Sustainability and LCA	Sustainability meets the needs of the present without compromising future generations.
Green Building Technologies	Green buildings minimize environmental impact through efficient use of energy, water, and materials.
Carbon Trading and RECs	Carbon trading allows organizations to buy and sell 'Carbon Credits' to meet emission targets.
International Governance and SDGs	International treaties like the Paris Agreement set global emission targets.

LEARNING RESOURCES

References

Recommended energy management references

1. Wayne C. Turner and Steve Doty, Energy Management Handbook.
2. B. L. Capehart, W. C. Turner and W. J. Kennedy, Guide to Energy Management.
3. Charles J. Kibert, Sustainable Construction: Green Building Design and Delivery.
4. United Nations, Sustainable Development Goals Report.

Approved public energy management sources

- https://nptel.ac.in/courses/112106318
- https://onlinecourses.nptel.ac.in/noc23_hs57/preview

These sources provide the verified study basis while official Course 6034B detail is unavailable; they do not replace official energy policies, authorised building assessments, certified utility plans or authorised sustainability goals.